CHAPTER

1

Medicine East and West: Two Ways of Seeing, Two Ways of Thinking

A story is told in China about a peasant who had worked as a maintenance man in a newly established Western missionary hospital. When he retired to his remote home village, he took with him some hypodermic needles and lots of antibiotics. He put up a shingle, and whenever someone came to him with a fever, he injected the patient with the wonder drugs. A remarkable percentage of these people got well, despite the fact that this practitioner of Western medicine knew next to nothing about what he was doing. In the West today, much of what passes for Chinese medicine is not very different from the so-called Western medicine practiced by this Chinese peasant. Out of a complex medical system, only the bare essentials of acupuncture technique have reached the West. Patients often get well from such treatment because acupuncture, like Western antibiotics, is strong medicine. But the theoretical depth and full clinical potential of Chinese medicine remain virtually unknown.

As a result, many Westerners have strange notions about Chinese medicine. Some of them see it as hocus-pocus—the product of primitive or magical thinking. If a patient is cured by

means of herbs or acupuncture, they see only two possible explanations: Either the cure was psychosomatic, or it was an accident, the happy result of hit-or-miss pin-sticking which the practitioner did not understand. They assume that current Western science and medicine have a unique handle on Truth—all else is superstition.

Other Westerners have a more favorable but equally erroneous view of Chinese medicine. Deeply and often justifiably disturbed by many of the products of Western science and culture, they assume that the Chinese system, because it is felt to be more ancient, more spiritual, or more holistic, is also more "true" than Western medicine. This attitude threatens to turn Chinese medicine from a rational body of knowledge into a religious faith system. Both attitudes mystify the subject—one by arrogantly undervaluing it, the other by setting it on a pedestal. Both are barriers to understanding.

Actually, Chinese medicine is a coherent and independent system of thought and practice that has been developed over two millennia. Based on ancient texts, it is the result of a continuous process of critical thinking, as well as extensive clinical observation and testing. It represents a thorough formulation and reformulation of material by respected clinicians and theoreticians. It is also, however, rooted in the philosophy, logic, sensibility, and habits of a civilization entirely foreign to our own. It has therefore developed its own perception of the body and of health and disease.

Chinese medicine considers important certain aspects of the human body that are not significant to Western medicine. At the same time, Western medicine observes and can describe aspects of the human body that are insignificant or not perceptible to Chinese medicine. For instance, Chinese medical theory does not have the concept of a nervous system. Nevertheless, it has been demonstrated that Chinese medicine can be used to treat neurological disorders.[1] Similarly, Chinese medicine does not perceive an endocrine system, yet it treats what Western

medicine calls endocrine disorders.[2] Nor does traditional Chinese medicine recognize *Streptococcus pneumoniae* as a pathological cause of pneumonia, yet often it effectively treats the disease.[3]

Chinese medicine also uses terminology that is strange to the Western ear. For example, the Chinese refer to certain diseases as being generated by "Dampness," "Heat," or "Wind." Modern Western medicine does not recognize Dampness, yet can treat what Chinese medicine describes as Dampness of the Spleen. Modern Western medicine does not speak of Fire, but can, from a Chinese perspective, stoke the Fire of the Kidney or extinguish excess Fire raging out of control in the Lungs. In Western medicine, Wind is not considered a disease factor; yet Western medicine is able to prevent Liver Wind from going to the head, or to extinguish rampaging Wind in the skin. The perceptions of the two traditions reflect two different worlds, but both can heal the same body.

The difference between the two medicines, however, is greater than that between their descriptive language. The actual logical structure underlying the methodology, the habitual mental operations that guide the physician's clinical insight and critical judgment, differs radically in the two traditions. What Michel Foucault says about medical perception in different historical periods could apply as well to these different cultural traditions: "Not only the names of diseases, not only the grouping of systems were not the same; but the fundamental perceptual codes that were applied to patients' bodies, the field of objects to which observation addressed itself, the surfaces and depths traversed by the doctor's gaze, the whole system of orientation of his gaze also varied."[4]

The two different logical structures have pointed the two medicines in different directions. Western medicine is concerned mainly with isolable disease categories or agents of disease, which it zeroes in on, isolates, and tries to change, control, or destroy. The Western physician starts with a symptom, then searches for the underlying mechanism—a precise *cause* for a

specific *disease*.[5] The disease may affect various parts of the body, but it is a relatively well-defined, self-contained phenomenon. Precise diagnosis frames an exact, quantifiable description of a narrow area. The physician's logic is analytic—cutting through the accumulation of bodily phenomena like a surgeon's scalpel to isolate one single entity or cause.

The Chinese physician, in contrast, directs his or her attention to the complete physiological and psychological individual. All relevant information, including the symptom as well as the patient's other general characteristics, is gathered and woven together until it forms what Chinese medicine calls a "pattern of disharmony." This pattern of disharmony describes a situation of "imbalance" in a patient's body. Oriental diagnostic technique does not turn up a specific disease entity or a precise cause, but renders an almost poetic, yet workable, description of a whole person. The question of cause and effect is always secondary to the overall pattern. One does not ask, "What *X* is causing *Y*?" but rather, "What is the relationship between *X* and *Y*?" The Chinese are interested in discerning the relationships among bodily events occurring at the same time. The logic of Chinese medicine is organismic or synthetic, attempting to organize symptoms and signs into understandable configurations. The total configurations, the patterns of disharmony, provide the framework for treatment. The therapy then attempts to bring the configuration into balance, to restore harmony to the individual.

This difference between Western and Eastern perception can be illustrated by portions of recent clinical studies done in hospitals in China.[6] In a typical study a Western physician, using upper-gastrointestinal X-rays or endoscopy by means of a fiberscope, diagnoses six patients with stomach pain as having peptic ulcer disease. From the Western doctor's perspective, based on the analytic tendency to narrow diagnosis to an underlying entity, all these patients suffer from the same disorder. The physician then sends the patients to a Chinese doctor for examination. The following results are found.

Upon questioning and examining the first patient, the Chinese physician finds pain that increases at touch (by palpation) but diminishes with the application of cold compresses. The patient has a robust constitution, a reddish complexion, and a full, deep voice. He seems assertive and even aggressive. He is constipated and has dark yellow urine. His tongue has a greasy yellow coating; his pulse is "full" and "wiry." The Oriental physician characterizes this patient as having the pattern of disharmony called "Damp Heat Affecting the Spleen."

When the Chinese physician examines the second patient, he finds a different set of signs, which comprise another overall pattern. The patient is thin. Her complexion is ashen, though her cheeks are ruddy. She is constantly thirsty, her palms are sweaty, and she has a tendency toward constipation, insomnia, and night sweats. She seems nervous and fidgety. Her tongue is dry and slightly red, with no "moss"; her pulse is "thin" and also a bit "fast." This patient is said to have the pattern of "Deficient Yin Affecting the Stomach," a disharmony very different from that of the first patient. Accordingly, a different treatment would be prescribed.

The third patient reports that massage and heat somewhat alleviate his pain, which is experienced as a minor but persistent discomfort. He is temporarily relieved by eating. The patient fears cold, has a pale face, sweats spontaneously in the daytime, and wants to sleep a lot. His urine is clear and his urination frequent; sometimes he has to get up in the middle of the night to empty his bladder. He appears timid, almost afraid. His tongue is moist and pale, his pulse "empty." The patient's condition is diagnosed as the pattern of "Exhausted Fire of the Middle Burner," sometimes called "Deficient Cold Affecting the Spleen."

The fourth patient complains of very severe cramping pain; his movement and affect is ponderous and heavy. Hot-water bottles relieve the pain, but massaging the abdomen makes it worse. The patient has a bright white face and a tendency toward loose stools. His tongue has an especially thick, white,

moist coating; his pulse is "tight" and "slippery." These signs lead to a diagnosis of the pattern of "Excess Cold Dampness Affecting the Spleen and Stomach."

The fifth patient experiences much sour belching and has headaches. Her pain is sharp, and although massaging the abdomen makes it diminish, heat and cold have no effect. She is very moody. Emotional distress, especially anger or melancholy, seems to precipitate attacks of pain; the pain is also worse during menses. Strangely enough, the patient's tongue is normal, but her pulse is particularly "wiry." The physician concludes that she is affected by the pattern of "disharmony of the Liver invading the Spleen."

The sixth patient has an extremely severe stabbing pain in the stomach that sometimes goes around to his back. The pain is much worse after eating and is aggravated by the slightest touch. He has had episodes of vomiting blood, and produces blackish stools. The patient is very thin and has a rather dark complexion. His tongue is a darkish purple and has markedly red eruptions on the sides. His pulse is "choppy." The Chinese physician describes the patient's problem as a "disharmony of Congealed Blood in the Stomach."

So the Chinese doctor, searching for and organizing signs and symptoms that a Western doctor might never heed, distinguishes six patterns of disharmony where Western medicine perceives only one disease. The patterns of disharmony are similar to what the West calls diseases in that their discovery tells the physician how to prescribe treatment. But they are different from diseases because they cannot be isolated from the patient in whom they occur. To Western medicine, understanding an illness means uncovering a distinct entity that is separate from the patient's being; to Chinese medicine, understanding means perceiving the relationships between all the patient's signs and symptoms. When confronted by a patient with stomach pain, the Western physician must look beyond the screen of symptoms for an underlying pathological mechanism—a peptic ulcer in this case, but it could have been an infection or a tumor or a nervous disorder. A Chinese physician examining the same pa-

tient must discern a pattern of disharmony made up of the entire accumulation of symptoms and signs.*

The Chinese method is thus holistic, based on the idea that no single part can be understood except in its relation to the whole. A symptom, therefore, is not traced back to a cause, but is looked at as a part of a totality. If a person has a symptom, Chinese medicine wants to know how the symptom fits into the patient's entire bodily pattern. A person who is well, or "in harmony," has no distressing symptoms and expresses mental, physical, and spiritual balance. When that person is ill, the symptom is only one part of a complete bodily imbalance that can be seen in other aspects of his or her life and behavior. Understanding that overall pattern, with the symptom as part of it, is the challenge of Chinese medicine. The Chinese system is not less logical than the Western, just less analytical.[7]

Yin (陰) and Yang (陽) Theory†

The logic underlying Chinese medical theory—a logic which assumes that a part can be understood only in its relation to the whole—can also be called synthetic or dialectical. In Chinese early naturalist and Taoist thought, this dialectical logic that explains relationships, patterns, and change is called Yin-Yang theory.††

*From a Western medical viewpoint, the Chinese physician is assessing the patient's specific and general physiological and psychological response to a disease entity.

†The Chinese characters for Yin and Yang are written here in the old style. Other characters in this book are simplified modern characters.

††Although the Chinese identify the relationships between phenomena primarily by the patterns of Yin and Yang, another system of categorization, known as the Five Phases, was also in use in early China. In this system, Wood, Fire, Earth, Metal, and Water were seen as a set of emblems by which all things and events in the universe could be organized. Although the Five Phases categories permeate virtually every aspect of traditional Chinese thought, leaving a significant impression on Chinese medical theory, this influence is for the most part formal and linguistic in nature. The Five Phases proved too mechanical, while Yin-Yang theory, because of its greater flexibility, was much more practical for the Chinese physician. It accommodated clinical changes and theoretical developments that the tradition required in order to grow. (For a detailed discussion of the Five Phases in Chinese medicine, see Appendix H.)

Yin-Yang theory is based on the philosophical construct of two polar complements, called Yin and Yang. These complementary opposites are neither forces nor material entities. Nor are they mythical concepts that transcend rationality. Rather, they are convenient labels used to describe how things function in relation to each other and to the universe. They are used to explain the continuous process of natural change. But Yin and Yang are not only a set of correspondences; they also represent a way of thinking. In this system of thought, all things are seen as parts of a whole. No entity can ever be isolated from its relationship to other entities; no thing can exist in and of itself. There are no absolutes. Yin and Yang must, necessarily, contain within themselves the possibility of opposition and change.

The character for Yin originally meant the shady side of a slope. It is associated with such qualities as cold, rest, responsiveness, passivity, darkness, interiority, downwardness, inwardness, and decrease.

The original meaning of Yang was the sunny side of a slope. The term implies brightness and is part of one common Chinese expression for the sun. Yang is associated with qualities such as heat, stimulation, movement, activity, excitement, vigor, light, exteriority, upwardness, outwardness, and increase.

Working with these ideas, Chinese thought and Chinese medical tradition have developed five principles of Yin and Yang.[8]

All things have two aspects: a Yin aspect and a Yang aspect.

Thus, time can be divided into night and day, place into earth and heaven, season into inactive periods (fall and winter) and active periods (spring and summer), species into female and male, temperature into cold and hot, weight into light and heavy, and so on. Inside and outside, down and up, passive and active, empty and full are all examples of Yin-Yang categories. These qualities are opposites, yet they describe relative aspects of the same phenomena. Yin and Yang qualities exist in relation to each other.

In terms of the body, the front is considered Yin and the back Yang. The upper part of the body is considered more Yang than the lower part; the outer parts of the body (skin, hair, etc.) are more Yang than the inner Organs. The Yin and Yang of the body are often described metaphorically as the body's Water and Fire. Illnesses that are characterized by weakness, slowness, coldness, and underactivity are Yin; illnesses that manifest strength, forceful movements, heat, and overactivity are Yang.

The philosopher Zou Yen (c. 305–240 B.C.E.) describes this idea this way: "Heaven is high, the earth is low, and thus [Heaven and Earth] are fixed. As the high and low are thus made clear, the honorable and humble have their place accordingly. As activity and tranquillity have their constancy, the strong and the weak are thus differentiated. . . . Cold and hot season take their turn. . . . [Heaven] knows the great beginning, and [Earth] acts to bring things to completion. . . . [Heaven] is Yang and [Earth] is Yin."[9]

Any Yin or Yang aspect can be further
divided into Yin and Yang.

This means that within each Yin and Yang category, another Yin and Yang category can be distinguished. It is an extension of the logic that divides all phenomena into Yin and Yang aspects, allowing further division within aspects *ad infinitum*. For example, temperature can be divided into cold (Yin) and hot (Yang), but cold can be divided further into icy cold (Yin) and moderately cold (Yang). In the body, the front of the trunk is Yin compared with the back, but the front can be divided further so that the abdomen is Yin in relation to the chest. Within a Yin illness characterized by Coldness there may be aspects of Yang such as sharp, forceful contractions. Within a Yang illness of Heat and hyperactivity there may be weakness and loss of weight, both Yin qualities.

Chuang Tzu (Zhuang Zi), the Taoist philosopher, (fl. probably between 400 and 300 B.C.E.) describes the unfolding of Yin and Yang, and the notion of the unity of opposites, in a radical paradoxical way: "There is nothing in the world greater than

the tip of a hair that grows in the autumn, while Mount Tai is small. No one lives a longer life than a child who dies in infancy, but Peng Zu (who lived many hundred years) died prematurely."[10]

Yin and Yang mutually create each other.

Although Yin and Yang can be distinguished, they cannot be separated. They depend on each other for definition. And the things in which Yin and Yang are distinguished could not be defined without the existence of Yin and Yang qualities. For instance, one cannot speak of temperature apart from its Yin and Yang aspects, cold and heat. Similarly, one could not speak of height unless there were both tallness and shortness. Such opposite aspects depend on and define each other.

Another example might be the relationship between a couple in which one partner can be (relatively) passive only if the other partner is (relatively) aggressive, and vice versa. Passivity and aggression can be measured only in comparison with each other. The activity (Yang) of the body is nourished by its physical form (Yin), and the physical form is created and maintained by the activity of the body. In illness, overactivity has meaning only in relation to a condition of underactivity, and vice versa.

Lao Tzu (Lao Zi), the reputed founder of Taoism, declares in the *Tao-te Ching* (or *Dao-de Jing*—the Classic of the Tao and Its Virtue):

> *Being and non-being produce each other;*
> *Difficult and easy complete each other;*
> *Long and short contrast each other;*
> *High and low distinguish each other;*
> *Sound and voice harmonize each other;*
> *Front and back follow each other.*[11]

Yin and Yang control each other.

If Yin is excessive, then Yang will be too weak, and vice versa. If the temperature is neither too cold nor too hot, then both cold and hot aspects are mutually controlled and held in check.

If it is too cold, then there is not enough heat, and vice versa. Yin and Yang balance each other.

In our example of the couple, the extent to which one partner can be aggressive depends on the extent to which the other is passive, and vice versa. They exert mutual control over each other. An illness of Fire in the body may be due to insufficient Water; an illness of Water may be due to insufficient Fire.

Lao Tzu alludes to this concept when he says:

> *He who stands on tiptoe is not steady.*
> *He who strides forward does not go.*
> *He who shows himself is not luminous.*
> *He who justifies himself is not prominent.*
> *He who boasts of himself is not given credit.*
> *He who brags does not endure for long.*[12]

Yin and Yang transform into each other.

This principle is a formula for the nature of organic process. It suggests two types of transformations: changes that occur harmoniously, in the normal course of events, and the sudden ruptures and transformations characteristic of extremely disharmonious situations.

Because Yin and Yang create each other in even the most stable relationships, Yin and Yang are always subtly transforming into each other. This constant transformation is the source of all change. It is a give-and-take relationship that is life activity itself. In the dynamics of the body, the nature of transformation can be illustrated by the manner in which inhalation is followed by exhalation, or periods of activity and exertion must be succeeded by nourishment and rest. In normal life such regular transformations occur smoothly, maintaining a proper, healthy balance of Yin and Yang in the body.

In a relationship in which Yin and Yang are unbalanced for prolonged periods of time or in an extreme manner, the resulting transformations may be quite drastic. Harmony means that the proportions of Yin and Yang are relatively balanced; disharmony means that the proportions are unequal and there is im-

balance. A deficiency of one aspect implies an excess of the other. Extreme disharmony means that the deficiency of one aspect cannot continue to support the excess of another aspect. The resulting change may be rebalancing or, if that is not possible, either the transformation into opposites or the cessation of existence.

To return to the couple, let's assume a disharmonious relationship in which one partner is excessively aggressive and the other excessively passive. This situation can have three possible outcomes: Either they sit down and talk it out, agreeing to a rearrangement of attitudes (i.e., they rebalance their relationship); or one day the passive partner gets fed up and waits for the other with an ax (i.e., a radical transformation of Yin into Yang occurs); or they separate, putting an end to the relationship.

In clinical practice, one of these three kinds of transformation is always possible. For example, when a patient has a pattern with very high fever and much sweating (considered an excess of Yang, or Fire), the patient may be in danger of suddenly going into shock (an extreme Yin, or Cold, condition). This is because Yang cannot continue to exist in such extreme relation to Yin without some transformation occurring. Either a gradual transformation, a rebalancing, must take place—medication and healing; or a radical transformation will occur—shock; or Yin and Yang will separate and existence will cease—death.

Lao Tzu describes the transformation process poetically:

In order to contract,
It is necessary first to expand.
In order to weaken,
It is necessary first to strengthen.
In order to destroy,
It is necessary first to promote.
In order to grasp,
It is necessary first to give.[13]

And also:

People hate to be orphaned, the lonely ones, and the unworthy.
And yet kings and lords call themselves by these names.
Therefore it is often the case that things gain by losing and lose by gaining.[14]

Yin-Yang theory is well illustrated by the traditional Chinese Taoist symbol (Figure 1). The circle representing the whole is divided into Yin (black) and Yang (white).

The small circles of opposite shading illustrate that within the Yin there is Yang and vice versa. The dynamic curve dividing them indicates that Yin and Yang are continuously merging. Thus Yin and Yang create each other, control each other, and transform into each other.

FIGURE 1

Traditional Yin-Yang Symbol



Because of the pervasive influence of Yin-Yang theory on Chinese thought and culture, the Chinese understand and explain events differently than does the West. The idea of causation, central to Western thinking, is almost entirely absent. Aristotle (384–322 B.C.E.), in his *Physics* (one of the basic works of

Western philosophy), pens the archetypal formulation of this Western notion: "Men do not think they know a thing till they have grasped the 'why' of it (which is to grasp its primary cause)."[15] For the Chinese however, phenomena occur independently of an external act of creation, and there is no great need to search for a cause.

> *Tao produced the One.*
> *The One produced the two.*
> *The two produced the three.*
> *And the three produced the ten thousand things.*
> *The ten thousand things carry the Yin and*
> *embrace the Yang and through the blending*
> *of the Qi* they achieve harmony.*[16]

In Chinese thought, events and phenomena unfold through a kind of spontaneous cooperation, an inner dynamic in the nature of things. Wang Cong (c. 27–100 C.E.), the great Taoist scientist, philosopher, and skeptic, describes the inner working of the universe as follows:

> The way to Heaven is to take no action. Therefore in the spring it does not act to start life, in summer it does not act to help grow, in autumn it does not act to bring maturity, and in winter it does not act to store up. When the . . . Yang comes forth itself, things naturally come to life and grow. When the . . . Yin arises of itself, things naturally mature and are stored up. . . . Originally no result is sought, and yet results are achieved. . . .
>
> Since Heaven takes no action, it does not speak. When the time comes for calamities and strange transformations, the [Qi] produces them spontaneously. . . . When there is [Cold] in the Stomach, it aches. It is not that man causes it. Rather, the [Qi] does it spontaneously. . . .[17]

Joseph Needham, the great historian of Chinese science,

*Qi, often spelled *ch'i* (or *ki* in Japanese), is discussed in Chapter 2.

summarizes the Chinese view of causation this way: "Conceptions are not subsumed under one another but placed side by side in a *pattern*, and things influence one another not by acts of mechanical causation, but by a kind of 'inductance'. . . . The key-word in Chinese thought is *Order* and above all *Pattern*. . . . Things behave in particular ways not necessarily because of prior actions or impulsions of other things, but because their position in the ever-moving cyclical universe was such that they were endowed with intrinsic natures which made that behavior inevitable for them. . . . They were thus parts in existential dependence upon the whole world-organism."[18]

The Chinese assume that the universe is continuously changing. Its movement is the result not of a first cause or creator, but of an inner dynamic of cyclical patterns. Just as the sun maps out four distinct seasons in its yearly round, so all biological organisms go through four seasons in a lifetime: birth, maturation, decline, and death. The constancy of the cosmos is in these patterns of change, which are regular. The cosmos itself is an integral whole, a web of interrelated things and events. Within this web of relationships and change, any entity can be defined only by its function, and has significance only as part of the whole pattern.

This metaphysics that emphasizes the perception of patterns is basic to Chinese thinking. It results in part from Taoism, which altogether lacks the idea of a creator, and whose concern is insight into the web of phenomena, not the weaver. For the Chinese, that web has no weaver, no creator; in the West the final concern is always the creator or cause and the phenomenon is merely its reflection. The Western mind seeks to discover and encounter what is beyond, behind, or the cause of phenomena. In the Chinese view, the truth of things is immanent; in the Western, truth is transcendent. Knowledge, within the Chinese framework, consists in the accurate perception of the inner movement of the web of phenomena. The desire for knowledge is the desire to understand the interrelationships or patterns within that web, and to become attuned to the unseen dynamic.

Of Chinese Painters and Chinese Physicians

The arts in China were nourished by the same naturalist and Taoist thought that fed Chinese philosophy and medicine. All Chinese art attempts to express the ideas of balance, harmony, and change that are contained in Yin-Yang theory. In landscape painting, the harmonious spirit of nature is revealed through the depiction of a scene that depends on proportion and measure to create beauty.

Wang Yun's *Mountain Landscape* (Figure 2) is a traditional Chinese landscape painting. In it, the artist has captured the essence of nature as he sees it, in balance and in flux. The painting is like the Taoist symbol (Figure 1), containing Yin and Yang in their proper proportions but constantly interacting and transforming into each other.

The scene depicts a vast range of elements, from the towering mountain to the little trickling stream. Nature is shown as a balance of the yielding Yin (foliage, water) and the unyielding Yang (rock, trees). There are the dynamic (water, people) and the quiescent (mountains, houses); the slow (trees) and the fast (mist); the dark and the light; the solid and the liquid. All things contain both Yin and Yang. The water, for instance, is both yielding (Yin) and dynamic (Yang).

The picture is a totality, and each detail takes on meaning only insofar as it participates in the whole. The mountain is immense by virtue of its smaller foothills; the people are small by virtue of the vastness of nature. All things are shown in their proper relationships to the things around them.

Through a kind of poetic logic, the Chinese also think of the landscape painting as a microcosm of the universe. In it are contained all the elements of nature, and it serves as a model of the cosmic process. The landscape painter sees in a particular scene a unique configuration of the natural elements. His composition includes those elements, or signs, that are specific to a certain scene, but it also participates in the larger reality. The elements within the microcosm correspond to elements within the macrocosm. For example, winter is death, a budding tree is



FIGURE 2

***Mountain Landscape* by Wang Yun (Qing dynasty)**

Courtesy Museum of Fine Arts, Boston;
purchased in China, received December 1913.

spring, a lake is all water, a person is humankind. The painting depicts a time and place that through their correspondence with the universe become timeless and placeless.

In a similar way, the Chinese think of each person as a cosmos in miniature. A person manifests the same patterns as does the painting or the universe. The Yang or Fire aspects of the body are the dynamic and transforming, while the Yin or Water aspects are the more yielding and nourishing. One person projects the heat and quickness of summer Fire; another person resembles the quiescence and coolness of winter Cold; a third replicates heaviness and moisture of Dampness; a fourth has the shriveled appearance of a Dry Chinese autumn; and many people display some aspects of the various seasons simultaneously. Harmony and health are the balanced interplay of these tendencies.

In each person, as in every landscape, there are signs that, when balanced, define health or beauty. If the signs are out of balance, the person is ill or the painting is ugly. So the Chinese physician looks at a patient the way a painter looks at a landscape—as a particular arrangement of signs in which the essence of the whole can be seen. The body's signs, of course, are somewhat different from nature's signs—including color of face, expression of emotions, sensations of comfort or pain, quality of pulse—but they express the essence of the bodily landscape.

Is Chinese medicine an art? Is it a science? If we mean by science the relatively recent intellectual and technological development in the West, Chinese medicine is not scientific. It is instead a prescientific tradition that has survived into the modern age and remains another way of doing things. But it does resemble science in that it is grounded in conscientious observation of phenomena, guided by a rational, logically consistent, and communicable thought process. It has a body of knowledge with standards of measurement that allow practitioners systematically to describe, diagnose, and treat illness. Its measurements, however, are not the linear yardsticks of weight, number, time, and volume used by modern science but rather images of

the macrocosm. Yet it also demands the artistic sensitivity of synthetic logic—always aware that the whole defines the parts and that the pattern may transform the significance of any one measurement within it: What is Yin in one person may be Yang in another. Because it deals with images, Chinese medicine allows and demands a recognition and assessment of quality.

This artistic sensitivity allows the physician to stay in touch with subtle refinements of meaning, to discern shades of significance in bodily signs; but most important, it allows awareness of the process that exists around and between linear measurements. Chinese medicine is not primarily quantitative. It recognizes that each person's pattern has a unique texture; each image has an essential quality.

The Chinese doctor's effort to recognize patterns within the characteristic signs of a particular individual is a creative one. But it is here that the concerns of the artist and the doctor begin to diverge. The artist uses his vision and skill to portray an ideal of balance and harmony on a paper scroll. The physician, however, uses his perception to recognize *dis*harmony, and must then apply his specialized skill to try to restore health—to achieve balance and harmony within a living organism.

Hallowed Tradition and Modern Research: It May Be Beautiful, but Does It Work?

Traditional medicine can be considered an art, and it can claim to be a science. But the important question to ask about a medical practice is: Does it work? Is Chinese medicine just an interesting philosophical curiosity or is it a viable system of healing? Can it treat what the West defines as real diseases? And can Western science measure its results and appreciate its value?

Because of the unique history of modern China, traditional medicine has been the subject of comprehensive study and testing over the past thirty years.

After the victory of the Chinese Revolution in 1949, the Chinese decided to take a fresh look at their traditional medical system. Many of China's new leaders were tempted to discard

their prescientific medical inheritance, along with other old-fashioned practices and remnants of underdevelopment. Their overall desire was to emulate the developed countries—to industrialize, electrify, and modernize. Another faction of the leadership, however, saw that although China did need to accept modern medicine, there might also be some practical and theoretical usefulness in the traditional medicine. The issue was whether or not it would prove efficacious from a modern perspective.[19] To answer that question, the Chinese performed thousands of experiments and clinical studies during the fifties.[20] The result was that in 1958 the Central Committee decided to give traditional and modern medicine equal respect and place in China.

The medical reports are still produced incessantly, and include such titles as the following.

Clinical Analysis of 290 Cases of Chronic Glomerular Nephritis Treated with Traditional Herbs.[21]

Observations of the Efficacy of Subcutaneous Acupuncture in Treating 121 Cases of Bronchial Asthma.[22]

Study of Treatment of Early Stage Cervical Carcinoma with Traditional Chinese Herbs, Including Analysis of Treatment Effect on Twenty-four Cases and a Preliminary Investigation of the Treatment Mechanism.[23]

Traditional Chinese Medical Treatment of Angina Pectoris—Report of 112 Cases.[24]

The pages of such studies fill entire libraries, yet it is not their quantity that is important, but rather their conclusions: that traditional Chinese medicine can hold its own, that it does work clinically.[25]

It is now evident that Chinese medicine is an effective healing method. Sometimes, traditional Chinese medicine can alleviate or treat illnesses that modern medicine is incapable of dealing with; at other times, the opposite is true, especially in those

cases that require surgery or intervention with high-technology equipment.[26]

The efficacy of traditional Chinese medicine can be illustrated by means of the previously mentioned study of ulcer patients. In that study, there were actually sixty-five patients, all of whom were observed by Western-trained physicians while being treated with traditional Chinese methods.[27] Every patient received a complete Western medical examination and was diagnosed as having peptic ulcer disease. The patients were then sent to traditional physicians, whose diagnoses roughly corresponded to the six different patterns described earlier.

The Chinese doctors gave each patient a unique herbal treatment based on his or her particular diagnosis. No Western treatment was administered, nor were any dietary restrictions imposed. The average length of treatment was two months, and modern Western techniques were used to evaluate the treatment's utility. The results were complete recovery in fifty-three patients (81.5 percent), significant improvement in seven patients (10.8 percent), some improvement in two patients (3.1 percent), no change in two patients, and a worsening of the condition in one patient due to complications unrelated to the treatment.

Another example is the angina study listed above. For that study 112 patients with angina pectoris were treated and observed for six months to two and a half years. All the patients were first given complete Western medical examinations and then diagnosed by traditional Chinese methods. In general, one of five distinct patterns was found in each case, and different herbs were prescribed for each pattern. Assessment of the therapy, based on subjective patient findings, was as follows: 34.8 percent of the patients improved markedly, 56.2 percent showed general improvement, and 9 percent remained unchanged. The overall percentage of subjective improvement was 91 percent. There were 91 patients who had abnormal electrocardiograph reports before treatment. Of these, 15.4 percent showed marked improvement after treatment, 23.1 percent showed moderate improvement, 10.9 percent showed an in-

crease to abnormal reports, and 50.6 percent showed no change. The most interesting results, however, were seen when the patients' blood was examined. The serum cholesterol and triglyceride levels had dropped markedly in all cases after treatment. Western medicine considers low levels of these substances beneficial in reducing the threat of atherosclerotic heart disease. These changes occurred without dietary restrictions and without the use of modern drugs to lower the level of blood lipids before or after treatment.[28]

Research has also been done to isolate the effective components of particular Chinese medical treatments in dealing with diseases recognized by the West. For example, experiments have discovered the active ingredients of herbs and have tried to develop efficient anesthesia using acupuncture. Many studies have attempted to extract from Chinese medicine new, Western-style cures for malaria, hypertension, viral infections, cancer, and other diseases. Common types of studies include "Use of the Traditional Herb *Desmodium rubrum* (Lour) D.C. to treat Epidemic B. Encephalitis"; "Traditional Herb, *Siegesbeckia orientalis*, in Treatment of Hypertension"; and "Antibiotic Properties of Fourteen Herbs on Diphtheria Bacillus and Their Action on Diphtheria Toxin."

Such studies are aimed at separating out the effective components of Chinese medicine and introducing them into the framework of modern Western medicine. Much of value has indeed been incorporated into the Western medicine that is practiced in China today. Someday, these components may well appear in Western medicine as practiced in the Occident. Yet, although this knowledge, with its use of traditional herbs and acupuncture, has the veneer of Chinese medicine, the actual application and methodology are clearly Western in orientation. The theory of Yin and Yang and other traditional concepts are left behind. And while new and valuable techniques are learned, the extraction of the practice from its theory calls into question the need for the traditional framework.

Fortunately for its future, however, the results of the studies generally demonstrate that traditional Chinese medicine does

work best when left in the context of Chinese logic.[29] In most cases, pattern weaving based on Yin-Yang theory produced better clinical results than the mechanical application of Chinese remedies within a Western context.[30] The traditional medicine works not only because it has an effective arsenal of therapeutic devices, but also because it knows best how to use them. Western observation also confirms that the medicine is especially effective in the clinic. This is because the Chinese view of health and disease as inseparable from a specific person means that the treatment will be well tailored to that person. Such personal shaping seems to maximize the effectiveness of the therapies.

Western clinical studies of traditional Chinese medicine, by proving its practical efficacy, have helped it win its battle for survival in the twentieth century, and promise it a place in the future of medicine.

Ancient but Still Alive

Chinese medicine is more than two thousand years old. Yet over all that time, it has retained an aesthetic and pragmatic relevance for humankind today. Of course, any tradition remains vital only insofar as it allows itself to grow and develop. The Chinese tradition is no different. Based on ancient and revered texts, it has continued to discover itself anew.

The *Huang-di Nei-jing* or Inner Classic of the Yellow Emperor (hereafter referred to as the *Nei Jing*) is the source of all Chinese medical theory, the Chinese equivalent of the Hippocratic corpus. Compiled by unknown authors between 300 and 100 B.C.E., it is the oldest of the Chinese medical texts. The knowledge and theoretical formulations it contains are the basic medical ideas developed and elaborated by later thinkers.

The *Nei Jing* has been called the bible of Chinese medicine, and the rest of Chinese medicine can be compared to rabbinical exegesis or interpretation of doctrine by church fathers. Just as, in the Jewish tradition, later authorities needed to explain theoretical issues raised by the Torah, so Chinese commentators added glosses on the *Nei Jing* that elucidated or even amended

its seminal ideas. The Chinese medical tradition thus brings together folk remedies and the therapeutics of China's physician-literati who served the Imperial Court.[31] It synthesized the medicine of one dynasty and another, one place and another, one thinker and another. Every dynasty has produced practitioners equal in stature to Galen, Avicenna, or Paracelsus, and all of them have made important additions and revisions to the tradition.[32]

In China today, the primary textbooks used to train traditional doctors are contemporary interpretations and clarifications of Qing dynasty (1644–1911) formulas and commentaries. These books are, in turn, clarifications of Ming dynasty (1368–1644) reworkings, which are also reworkings of earlier material. This process goes all the way back to the Han dynasty (202 B.C.E.–220 C.E.). Such transmission through the dynastic pathway not only preserved and encapsulated the original sources, but also elucidated and reformed them. (See Appendix I: Historical Bibliography.)

It is for this reason that the *Nei Jing*, although it is the source of the tradition, is usually one of the last texts to be studied in contemporary schools of Chinese medicine. The *Nei Jing*, written in archaic language, is often unclear and inconsistent, and can only be understood after much preparation.[33] Without the commentaries and modifications of later eras, the *Nei Jing* would be almost completely unintelligible.[34] So the source requires the tradition to explain it, but both are necessary to guide Chinese theory and practice.

Within Chinese medicine, as in all traditional systems, there is a tension between that which is tacitly recognized as no longer useful and that which continues to be accepted as profound. This book attempts to bring Chinese medicine to a Western audience, and because it does so within the ancient tradition, it is, finally, another commentary on the commentaries.